

Week 7 Cost-Benefit Memo

Complex Models vs. the Week 6 Baseline: A Recommendation

Verdict: we recommend the tuned Random Forest for Phase 3 — not because it is the most accurate model overall, but because it more than doubles our ability to catch the most critical (ESI 1) patients, at a training cost measured in seconds rather than minutes, while still being explainable to a clinician in under a minute.

1. Recap: dataset and method

Same Week 6 dataset and split throughout — 55,121 ED visits, 80/20 stratified split, random_state=42 (44,096 train / 11,025 test) — reused unchanged so every comparison below is honest. Four models were benchmarked: the Week 6 Logistic Regression baseline, a tuned Random Forest, Gradient Boosting (HistGradientBoostingClassifier), and a small MLP (two hidden layers, 64 then 32 neurons). All models except the baseline used class_weight="balanced" or an equivalent mechanism to give the rare, critical ESI 1 class more weight during training.

2. Benchmark table all

Model	Accuracy	Precision	Recall	F1	Recall ESI1	Train	Infer/pred
Baseline (LogReg)	0.671	0.571	0.452	0.481	0.188	5.0s	0.003ms
Random Forest (tuned)	0.573	0.417	0.592	0.436	0.500	7.2s	0.027ms
Gradient Boosting	0.550	0.410	0.547	0.416	0.312	8.2s	0.008ms
Small MLP	0.638	0.526	0.482	0.498	0.312	103.0s	0.003ms

Precision, recall, and F1 are macro-averaged (every ESI level counted equally). Recall ESI1 is reported separately because it is the single number this project has repeatedly found to matter most clinically. Inference time is per single prediction; at this scale, every model is fast enough that latency is not a deciding factor — training cost and the two recall columns are what actually separate these models.

3. Where each model actually goes wrong (the ESI-1 row)

Aggregate metrics hide who gets hurt. Reading the ESI-1 row of each model's confusion matrix directly — of the 16 truly critical patients in the test set:

Model	Correctly caught (ESI1)	Downgraded to ESI2	Downgraded to ESI3+
Baseline (LogReg)	3 of 16	11	2
Random Forest (tuned)	8 of 16	6	2

The Random Forest more than doubles the raw catch count (3 → 8) and, critically, shrinks the worst error — patients downgraded all the way to ESI 2 ("emergent" instead of "resuscitate") — from 11 to 6. Every remaining miss is still a serious failure, not a solved problem, but it is a materially smaller one.

4. Three arguments for the Random Forest

- Best recall on both axes that matter clinically. Recall_macro (0.592) and recall on ESI 1 specifically (0.500) are both the highest of the four models — it catches half of the truly critical patients, nearly triple the baseline's 0.188.

- Cheap to train and to re-train. A full retrain costs about 7 seconds on modest hardware — trivial for Martina Griffith's team to redeploy on a schedule, versus the MLP's 103 seconds for a worse ESI-1 catch rate.
- Passes the one-minute interpretability test. `feature_importances_` plus a real patient example (a stroke-alert case with blood pressure 156/138, correctly flagged as ESI 1 despite a normal-looking shock index) gives a plain-language explanation Dr. Reyes can sanity-check without a data science background.

5. Three arguments against

- Lower overall accuracy and precision than the baseline (0.573 vs 0.671 accuracy; 0.417 vs 0.571 macro precision). It will raise more false alarms across the less-urgent classes — more patients get flagged as more urgent than they turn out to be, which has its own real cost to ED flow and staff trust.
- The hyperparameter search behind this model was deliberately narrowed for compute reasons during development (6 random configurations over 2-fold cross-validation, rather than a fuller search). The reported numbers may not be this model's ceiling — a wider search could shift the picture in either direction.
- Feature importances are a weaker form of interpretability than the baseline's literal coefficients. "These features mattered most, on average, across many trees" is a real explanation, but it is a step down in rigor from a logistic regression's directly readable, per-feature weights.

6. Why not Gradient Boosting or the MLP

Gradient Boosting: cheap to train (8.2s) and fast at inference, but it doesn't win anywhere that matters. Its ESI-1 recall (0.312) is well below the Random Forest's (0.500), and `HistGradientBoostingClassifier` has no built-in `feature_importances_` — explaining a single prediction requires extra tooling (permutation importance or SHAP) that hasn't been built or validated here. It loses on the clinical metric and on interpretability, for no compute advantage that matters at this scale.

Small MLP: the best accuracy (0.638) and macro-F1 (0.498) of the four models, and genuinely worth remembering for later phases — but for this recommendation it costs roughly 14x the Random Forest's training time (103s vs 7.2s) and ties Gradient Boosting's weaker ESI-1 recall (0.312 vs 0.500). It also has no straightforward interpretability story at all: there is no equivalent of feature importances or coefficients to point to. A model that costs more to train, catches fewer critical patients, and cannot be explained is difficult to justify to a clinical board, regardless of its aggregate accuracy.

7. For IT Governance (Martina Griffith)

Framed directly against compute and audit cost, since that's the concern raised at kickoff:

- Retraining cost: ~7 seconds on a single modest CPU core with no GPU requirement. This can run on a nightly or weekly schedule without dedicated infrastructure.
- Inference cost: ~0.03ms per prediction — trivial. At 11,025 predictions (a full day's test-set volume for scale), total inference time is a fraction of a second.
- Storage/dependency footprint: a single scikit-learn `RandomForestClassifier` object, no GPU, no specialised serving stack — the same Python environment already running the Week 6 baseline can serve this model.
- Auditability: `feature_importances_` ships with the model for free and was used directly in Section 4 to explain a real prediction — no additional tooling purchase or SHAP integration is required to meet a basic "can you explain this" bar, though a fuller audit tool is recommended before production sign-off.
- No hedging on the honest cost: this model will flag more patients as more urgent than the baseline does, which means more clinician time spent reviewing and clearing false alarms. That staffing cost is real and outside this memo's scope to quantify, but the ED Board should weigh it explicitly, not discover it after deployment.

8. Risks and unknowns

- No fairness or subgroup error audit yet — demographics were checked for a raw accuracy effect (negligible: -0.001 macro-F1) but not for whether errors cluster unevenly across age, ethnicity, or race groups.
- No external validation. Every number here comes from one hospital's historical data, split from the same distribution. Week 6's caution about a model that "memorised our data and may not travel to a new hospital" still applies.

- The recall gain comes from class-weighting, not a deliberately chosen decision threshold matched to the ED's actual tolerance for false alarms — that tuning step hasn't been done yet.

9. Recommendation

Proceed with the tuned Random Forest as the Phase 3 candidate, conditional on three follow-ups before any deployment conversation: (1) a fuller hyperparameter search once more compute is available, since the current tuning was constrained by development hardware, not by the model's real ceiling; (2) a subgroup error audit across demographic groups, which this project has not yet done for any model; and (3) explicit clinician-in-the-loop review of every Random-Forest-flagged case rather than autonomous triage action, given the accuracy trade-off documented above. On the numbers available today, this is the most defensible balance of catching critical patients, training cost, and the ability to explain a decision to the people who have to trust it.